SAHAYAK | SCIENCE EXHIBITION
SHYAMANTIKA PEGU
INTRODUCTION, PROBLEM & PURPOSE
ROLE-SPECIFIC SPEAKING SCRIPT
Suggested speaking time: 3-5 minutes

OPENING
Good morning respected judges, teachers, and everyone present here. We are presenting SAHAYAK, a real-time Indian Sign Language recognition system. I will introduce the communication challenge that inspired our project, explain its purpose, and describe what we aim to demonstrate.

THE PROBLEM
People who use sign language may face barriers when others around them do not understand signs. This can make everyday communication in classrooms, public services, and other settings more difficult. Sign languages are full languages with their own grammar and regional variation; they are not simply gestures or word-for-word versions of spoken language.

WHY SAHAYAK
We wanted to explore how computer vision and machine learning might help in a specific accessibility task. SAHAYAK uses a camera to detect hands, extracts hand landmarks, and predicts a label from a limited set of signs that its model has been trained to recognise. The website displays the prediction, and supporting speech or translation features may be available depending on connectivity.

SCOPE AND HONEST LIMITATIONS
SAHAYAK is a prototype, not a complete ISL translator. It does not understand every sign, sentence, facial expression, grammar rule, or regional variation. Lighting, camera angle, occlusion, and differences between signers can affect results. A broader system would need more representative data, systematic testing, and feedback from ISL users and experts.

HANDOVER
That explains the problem and our goal. My teammate will now explain how the system works technically, from camera input and hand landmarks to numerical features and model prediction.

JUDGE Q&A
Is sign language universal? No. Different sign languages exist and have their own linguistic rules.
Can it replace an interpreter? No. It is a limited prototype and should not replace qualified interpreters.
Why is it useful? It demonstrates one possible assistive-technology approach and helps identify what must improve before real-world use.

REMINDERS
Speak clearly, make eye contact, and do not claim unsupported accuracy or user counts. Explain that the project recognises only its trained labels. If asked about

an unmeasured result, say that it has not yet been measured.